

# A Prompt-Engineered LLM Framework for Extracting ESA Narratives with Narrative Policy Framework-Based Coding.

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## Abstract

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## Keywords

Narrative Policy Framework, Large language models

## 1 Introduction

Rapid biodiversity loss and growing extinction risk have made public understanding of conservation policy a critical component of environmental governance. In the United States, the Endangered Species Act (ESA) was passed by Congress in 1973 as the primary federal instrument for responding to this crisis [1]. The ESA is widely regarded as one of the most powerful and controversial conservation laws, designed to prevent extinction by listing species as threatened or endangered, protecting their habitats, and directing federal agencies to avoid actions that endanger listed species [2]. Yet despite its scientific and legal foundations, the ESA is rarely encountered by the public through direct engagement. Instead, most people come to understand the law through the stories told about it — stories about which species are at risk, who is responsible for that risk, and what should be done to address it. [3]. Wildlife policy conflicts involving species such as wolves and grizzly bears show this clearly: decisions about wildlife management often reflect deeper cultural and political differences, and the disagreement is usually about more than just facts [? ? ]. In Idaho, debates over wolf recovery show this especially well, as the conflict has involved not only ecological questions but also politics, identity, and competing views about management [? ]. These conflicts show that conservation issues are not only about science, but also about values, authority, and control. In that sense, environmental policy debates rarely turn on facts alone: the ESA is not only a legal and scientific framework but also a politically contested issue that is often framed through narratives about which species are at risk, who is responsible for that risk, and what should be done to address it rather than through policy facts alone.

This narrative quality of ESA debate is not incidental to the policy process, it is part of how the policy process operates. Public policy issues are often communicated through competing stories about who is responsible, what the problem is, and what should be done. This indirect relationship between the public and environmental policy elevates the role of media from a passive information channel to an active political actor. Media institutions translate complex scientific and legal information into accessible accounts, selectively emphasizing certain issues, actors, and interpretations while downplaying others. Through agenda-setting, the frequency and prominence of coverage shape which ESA issues and species are perceived as important, bringing some policy questions to the

forefront while leaving others largely invisible [4]. Through framing, media narratives organize these issues into coherent interpretations by defining problems, assigning causal responsibility, and highlighting particular moral evaluations [5]. In ESA discourse, this often appears as competing portrayals of conservation versus economic cost, or federal protection versus local control. Through priming, repeated exposure to these frames conditions the criteria that audiences use when evaluating policies and political actors, influencing whether the ESA is judged in terms of ecological necessity, regulatory burden, or ethical obligation [6]. Empirical work shows that ESA coverage frequently emphasizes conflict and controversy, suggesting that media do not simply increase attention but also structure how the policy is interpreted and judged [2, 7, 8]. In this sense, media do not merely report ESA debates, they actively construct the communicative environment in which those debates unfold [9].

The importance of this narrative struggle becomes even clearer when compared with public support for the ESA. Survey evidence shows that support for the law has remained high and relatively stable over time, even as political debate around the ESA remains intense and polarized [? ]. This creates an important puzzle: if public support is steady, why does ESA discourse remain so conflicted? One answer is that the policy debate is shaped not only by public preferences, but also by the narratives told by media, policymakers, and interest groups. Understanding how those narratives are formed, how they circulate, and how they influence interpretation is therefore essential to understanding endangered species politics itself.

Treating agenda-setting, framing, and priming as separate processes, however, can hide how policy communication actually works in practice. In domains like the ESA, these mechanisms are connected and operate together through broader narrative structures that link problems, actors, causes, and solutions into coherent stories. The Narrative Policy Framework (NPF) offers an integrative way to study this process by treating policy discourse as structured storytelling [? ? ]. Within the NPF, setting directs attention, characters are assigned roles such as heroes, villains, and victims, plots establish causal relationships, and policy solutions express normative judgments about what should happen next. This makes the NPF especially useful for ESA debates, where species are often portrayed as victims, regulators as protectors or overreaching authorities, and policy choices as moral decisions about conservation and economic trade-offs [10, 11]. Compared with approaches such as topic modeling, which capture word co-occurrence but not causal or evaluative meaning, the NPF offers a stronger way to understand how narratives structure policy conflict and communication [? ].

This paper builds on that logic by asking how ESA narratives are constructed, how they move through media and political institutions, and whether they shape policy outcomes. In policy research, narratives are not just descriptions of events; they are strategic tools that policymakers and other actors use to influence interpretations, guide behavior, and advance policy goals. As Gray and Jones (2016) note, narratives may be used directly or indirectly to shape the actions of relevant actors, while McBeth and Shanahan (2004) emphasize that policy narratives are intentionally constructed and purposive. This makes narrative analysis especially important for understanding how political meaning is produced, contested, and institutionalized.

In the context of ESA politics, narrative analysis helps identify the central actors, key arguments, conflicts, and anticipated consequences surrounding endangered species policy. It also allows the researcher to examine how different policy actors frame ESA issues, justify policy positions, and present scenarios of future action or inaction. This is consistent with broader policy narrative scholarship, which argues that narrative analysis can clarify a political issue, reveal the policy context, identify the main actors and arguments, and illuminate possible future developments [12? ?]. The importance of narrative research has also been recognized in studies of new technological and organizational phenomena, where narrative forms help explain complex and contested change [? ? ?].

This paper develops a prompt-engineered large language model framework for extracting ESA narratives using Narrative Policy Framework-based coding. The goal is not only to measure narratives at scale, but also to test whether a theory-driven LLM approach can capture the structure of ESA discourse more effectively than traditional text-as-data methods. Taken together, the paper argues that ESA politics cannot be understood through law or science alone; it must also be understood as a communicative and strategic policy process in which narratives organize attention, interpretation, and evaluation, and ultimately shape how the law is debated and implemented [? ?].

## 2 Related Work

### 2.1 The Role of Media in Shaping Policy Attention, Interpretation, and Evaluation

Scholars have long debated the role of media in shaping public policy, particularly how media narratives influence public opinion and create political pressures that affect legislative decision-making [13]. One central pathway runs through agenda setting and the ways issues are framed for specific policies, such that media attention and narrative emphasis help determine which problems become politically urgent and how they are understood by elites and citizens alike. This perspective builds on classic agenda-setting theory, which holds that mass media do not simply mirror public concerns but actively structure the public agenda by allocating scarce attention across competing issues [4]. When news organizations repeatedly highlight certain topics — crime, inflation, environmental crises — those topics become more salient in citizens’ minds, shaping perceptions of what government should prioritize and, in turn, creating incentives for legislators to defend or oppose particular policy responses. Sustained media coverage can elevate specific policy issues onto

the political agenda, influence the salience of attitudes toward those issues, and condition the space within which policy debates unfold [4, 14]. This claim is not merely theoretical. (author?) [4]’s original study operationalized agenda setting empirically, comparing the frequency of issue coverage during the 1968 U.S. presidential election against survey responses identifying the “most important problem” facing the country, and found a strong correlation between media issue salience and public issue importance — evidence that media attention shapes what citizens think about rather than what they think. Subsequent research has replicated and extended these findings across policy domains and political contexts [15, 16], though later studies show that agenda-setting effects are conditional rather than universal: they tend to be stronger for less noticeable or complex issues where citizens lack direct personal experience [17], and may weaken in fragmented media environments characterized by selective exposure [18]. Time-series analyses further suggest that the relationship between media and public salience may run in both directions, with public concern sometimes driving media coverage rather than the reverse [19] — a possibility directly relevant to this dissertation’s second chapter, which asks whether media lead congressional attention to the ESA, whether Congress leads media coverage, or whether the relationship is genuinely reciprocal.

Influencing issue salience, however, represents only one dimension of media power. Media coverage may not only determine which issues receive attention but also shape how those issues are evaluated and interpreted, meaning agenda setting is only one of several mechanisms through which media can shape policy-relevant attitudes and behavior [20]. Framing and priming operate as two further, related but distinct mechanisms. Where agenda-setting research asks which issues become salient, framing research asks how alternative descriptions of the same issue change interpretation. Iyengar’s experiments on episodic versus thematic television news show that different frames of the same social problem shift whether citizens attribute responsibility to individuals or to society [21], extending earlier psychological work on gain-loss framing in decision-making [22]. Subsequent studies show that news stories foregrounding different core values — public order versus free speech, for instance — alter tolerance for controversial groups [23], and that framing effects are strongest when the frame originates from a credible source [24]. More recent reviews emphasize that framing is highly conditional, weakening when competing frames are present, when individuals hold strong prior attitudes, or as time passes without reinforcement [25, 26]. Framing, in other words, is best understood not as a uniform media effect but as a context-dependent mechanism operating under specific informational and audience conditions — a description that maps closely onto ESA discourse, where the same listing decision can be framed alternately as ecological necessity, regulatory overreach, or economic threat depending on the outlet and the audience being addressed.

Priming, the third mechanism, was originally investigated in connection with agenda setting, grounded in the idea that repeated media attention to an issue increases its accessibility in memory and, in turn, influences the criteria citizens use to evaluate political actors and policies [27]. (author?) [28] extend this logic, arguing that the more attention media gives to a particular issue domain, the more

citizens are primed to incorporate what they know about that domain into broader political judgments — for instance, evaluations of presidential performance shifting depending on how heavily the economy has been covered. Yet priming evidence is mixed: a pre-registered replication under contemporary media conditions finds no average priming effect at all [29], while a meta-analysis concludes that priming is strongest in political news contexts and among less knowledgeable audiences, weakening as individuals are exposed to more diverse media environments [30]. Together, these findings suggest that priming is a genuine but fragile mechanism, dependent on sustained issue emphasis and particular audience conditions rather than operating automatically whenever coverage increases.

## 2.2 The Narrative Policy Framework

The previous subsection traced how agenda setting, framing, and priming operate as related but conceptually distinct media effects, and closed by noting that their boundaries are often blurred in practice because all three rely on overlapping textual indicators and overlapping cognitive processes of accessibility and applicability. This raises the question of whether the Narrative Policy Framework (NPF) offers unique analytical leverage over these alternative approaches, and how its core elements can generate testable hypotheses about policy dynamics in the ESA context that motivates this dissertation.

The NPF addresses this challenge by shifting analytical focus from isolated media effects to integrated narrative structures that simultaneously organize attention, interpretation, and evaluation through four structural components: setting (which directs agenda attention), characters (which prime relevant actors), plot (which frames causal relationships), and policy solution (which prescribes normative evaluation) [31, 32]. Rather than treating agenda setting, priming, and framing as separable processes that must each be operationalized and tested on their own, the NPF maps its elements directly onto these cognitive mechanisms — setting to issue salience, characters to decision weights, plot to causal interpretation, and solution to evaluation criteria — while providing a computationally structured framework for text analysis that enforces narrative logic within observable discourse units [31, 33]. This line of thinking raises the possibility that narrative-oriented frameworks such as the NPF [34] may offer useful conceptual tools for examining how accessibility and applicability interact within policy communication, without presuming that any single media effect operates in isolation.

The NPF also offers distinct advantages over alternative theoretical frameworks that might otherwise be applied to ESA discourse. The Multiple Streams Framework captures problem-stream attention through policy windows but neglects the character-driven causality central to framing effects [35]; Policy Feedback Theory traces institutional effects on preferences over time but misses strategic elite storytelling [36]; and standard topic models identify thematic clusters without enforcing narrative coherence [37? ]. The NPF uniquely synthesizes agenda setting, priming, and framing into a single structure, treating policy communication as storytelling rather

than as discrete shifts in issue salience or isolated frames [? ]. Settings, characters, and causal plots specify not only which issues appear on the agenda, but also who is responsible for problems and which evaluative standards should apply, thereby integrating all three media-effects mechanisms into one coherent narrative structure [38]. (author?) [31] introduce the NPF specifically as a quantitative, structuralist, and positivist approach that acknowledges the importance of narratives in policy processes and seeks to enable more systematic and productive empirical debate over how stories influence public policy. In this sense, the NPF can be read as an attempt to reconcile and extend classic media-effects models rather than to replace them.

This narrative perspective has been demonstrated productively outside the environmental domain as well. (author?) [39] illustrates the analytical value of the NPF in the study of economic communication by combining a backward-looking conception of inflation narratives with large language models, showing how the NPF can be used to systematically analyze media narratives as causal explanations of economic outcomes. The same underlying structure — settings, characters, plots, and moral or policy conclusions — applies just as naturally to ESA discourse, where species are routinely cast as victims, regulatory agencies alternately as protectors or overreaching authorities, and policy choices as moral judgments about conservation and economic trade-off [10, 11, 40].

Despite these conceptual strengths, however, scholars have increasingly noted that the NPF has been more successful as a descriptive framework than as a measurement framework. As several reviewers observe, most applications of the NPF have remained largely descriptive, relying on manual coding of a limited set of texts and offering few scalable tools for measuring narrative elements in large corpora [34]. This measurement gap is particularly acute when the research goal is to trace temporal feedback between media and policymaking venues, where narratives may originate in one arena and diffuse or transform as they move to another — precisely the question this dissertation’s second chapter takes up for ESA discourse between the press and Congress. A parallel observation arises in the study of economic narratives, where understanding behavior depends on the stories people tell, yet quantitative approaches to analyzing those narratives remain largely underdeveloped [41]. One of the central challenges identified in this literature is simply the difficulty of measuring narratives in written or spoken text [42]; any adequate measure must capture relationships between characters, concepts, organizations, or objects [43]. Traditional text-as-data methods in social science, which typically focus on word frequencies or broad topic patterns, fail to account for these relational structures, leaving the causal structure of narratives unmeasured [44, 45].

## 2.3 Computational and LLM-Based Approaches to Narrative and Policy Text

The previous subsection closed by noting that the Narrative Policy Framework has historically been more successful as a descriptive framework than as a measurement framework, with most applications relying on manual coding of a limited set of texts and offering

few scalable tools for extracting narrative elements from large corpora [34]. This subsection traces how computational text analysis methods have attempted to close that gap, moving from general-purpose text-as-data methods through structural topic modeling and entity-relation extraction to the recent emergence of large language models, and shows where each step still falls short of the goal this dissertation pursues: extracting the full structure of an NPF narrative setting, characters, causal plot, and moral or policy conclusion simultaneously, rather than modeling text along a single dimension such as topic or sentiment.

The starting point for this line of work is the broader text-as-data tradition in political science, which demonstrated the promise and the limits of automated text analysis well before narrative-specific methods existed [44]. General text-as-data methods, including topic models and supervised classification, made it possible to analyze political text at a scale no team of human coders could match, but these methods were built to surface themes and word patterns rather than the relational structure — who did what to whom, and why — that defines a narrative. [?] extended this tradition with Structural Topic Modeling (STM), which models latent themes in open-ended text at scale while accounting for document-level covariates through metadata-informed prevalence estimation. STM represented a substantial advance in scalability and remains a natural baseline for narrative-adjacent text analysis, including in this dissertation’s first empirical chapter, but it identifies thematic clusters without enforcing any narrative coherence across them [37]: a topic model can show that a corpus discusses, say, wolves, ranchers, and federal regulation in similar proportions, but it cannot say who is cast as the hero or villain of that story, nor what causal claim links those actors together.

Closing that relational gap required a different methodological move. (author?) [42] addressed the challenge of measuring narratives directly by developing RELATIO, an open-source Python package for measuring interpretable narratives from plain text through structured semantic relations between entities, applied originally to congressional speech. Unlike topic models, a RELATIO-style entity and relation extraction pipeline captures who is doing what to whom within a text, moving analysis closer to the causal and relational structure that NPF research requires [43]. (author?) [46] pushed this further by combining dynamic topic models with narrative-shift detection using temporal embedding alignment, allowing researchers to track how policy narratives shift over time in large text corpora rather than treating a corpus as a single static snapshot — a methodological step directly relevant to this dissertation’s interest in how ESA narratives move and transform between media and congressional venues over time.

Despite these advances, (author?) [39] identify a fundamental limitation that persists across this entire line of work: although dynamic topic models and entity-relation pipelines address scalability, they lack the integrated language comprehension required to fully capture the contextual meaning and relevance of a narrative as a whole. A relation-extraction model can recover that one entity acted on another, but it does not, on its own, understand whether that action constitutes a moral transgression, a policy solution, or an incidental

detail — distinctions that are central to the NPF’s structural elements. To address this gap, (author?) [39] propose an integrated approach that leverages large language models such as GPT [47] to identify economic narratives directly from large text corpora, treating the LLM not merely as a faster classifier but as a tool capable of the kind of contextual reading that narrative extraction actually requires. This approach is further supported by [?], who benchmark GPT-style models systematically across natural language processing tasks and demonstrate that large language models exhibit remarkable proficiency on tasks requiring deep comprehension of language and context, underscoring their potential to capture complex narrative structures that traditional NLP pipelines struggle to extract.

Table 1 summarizes this progression and the specific gap this dissertation addresses. Each step in this literature — from general text-as-data methods, through structural topic modeling, through entity-relation extraction, to large language model benchmarking and economic narrative extraction — expands what can be measured computationally, but no prior study applies these tools to extract a complete NPF narrative structure from policy text in a single, explicit, repeatable procedure. Topic models capture themes but miss narrative structure entirely; recent LLM-based approaches demonstrate strong language comprehension but are not yet paired with an explicit methodological design for extracting NPF components specifically. This dissertation’s first empirical chapter closes that gap by developing a prompt-engineered LLM pipeline that treats prompt design itself as an algorithm, extracting all four NPF elements — setting, characters, causal plot, and policy or moral conclusion — simultaneously from each document in the ESA corpus, and evaluating that pipeline directly against Structural Topic Modeling as a baseline rather than assuming the LLM-based approach is superior outright.

**Table 1: Prior work on computational narrative and policy text analysis, and the gap this dissertation addresses.**

| Study                      | Goal                                                                                                                       | Methods used                                                                                     | Reference      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------|
| Jones and McBeth (2010)    | Establish the Narrative Policy Framework and define the four structural elements of narrative in policy discourse.         | Conceptual NPF framework; no computational extraction.                                           | (author?) [48] |
| Grimmer and Stewart (2013) | Demonstrate the promise and limits of automated text analysis for political science research.                              | General text-as-data methods including topic models and supervised classification.               | (author?) [49] |
| Roberts et al. (2014)      | Model latent themes in open-ended text at scale while accounting for document-level covariates.                            | Structural topic modeling (STM) with metadata-informed prevalence estimation.                    | (author?) [50] |
| Ash et al. (2024)          | Measure interpretable narratives from plain text using structured semantic relations between entities.                     | RELATIO-style entity and relation extraction applied to congressional speech.                    | (author?) [51] |
| Shahriar et al. (2024)     | Evaluate whether large language models can perform deep language understanding and context-sensitive classification tasks. | Systematic benchmarking of GPT-style models across natural language processing tasks.            | (author?) [52] |
| Lange et al. (2025)        | Track how policy narratives shift over time in large text corpora.                                                         | Dynamic topic models combined with narrative-shift detection using temporal embedding alignment. | (author?) [53] |
| Schmidt et al. (2025)      | Identify economic narratives directly from large corpora without pre-specified category systems.                           | Integrated NLP pipeline combining entity extraction with large language model classification.    | (author?) [39] |

### 3 Methodology

#### 3.1 Datasets

We developed two text corpus consisting of news media articles and congressional documents. These two datasets capture complementary dimensions of policy discourse where news media represent the narratives presented to the public, while congressional records document how legislators frame, debate, and justify ESA-related policy decisions.

#### 3.2 Prompt-based Zero-Shot and Structured Narrative Extraction Approaches

**Zero-Shot Prompting.** Zero-shot prompting refers to asking a language model to perform a task using only instructions, without providing example input-output pairs. It is especially useful when the task is clearly defined but no labeled training examples are available [54? ?]. Because the model receives no demonstrations, zero-shot prompting places the full burden of task understanding on the natural language instruction itself, requiring the prompt designer to specify what to look for, what to return, and what to avoid with sufficient precision that the model can generalise the instruction to any input in the target domain [55]. In our case, zero-shot prompting serves as the core design principle of the extraction framework, but it is supplemented by additional extraction techniques that help structure the outputs and improve consistency across documents. The decision to use zero-shot rather than few-shot prompting across all three steps was deliberate and directly motivated by the multigenre nature of the corpus. Few-shot prompting in which a small number of labeled input-output examples are embedded in the prompt to demonstrate the task has been shown to produce reliable performance gains when the examples are drawn from the same distribution as the target inputs [56]. However, providing examples from only one genre can bias the model toward that genre’s style and conventions, which may reduce accuracy when the same prompt is applied to texts from other genres.[57, 58].

**Role-based prompting.** Together with the zero shot prompting, Role-based prompting, also referred to as persona prompting in the literature, assigns the model a specific identity or expert role such as an analyst, coder, or extractor to shape the style and content of its response, which helps steer the model toward the desired perspective and output behavior[59? ]. Rather than issuing a generic instruction, the prompt opens by telling the model what kind of expert it is, which has been shown to prime domain appropriate vocabulary and judgment without requiring worked examples, because the assigned role activates relevant prior knowledge already encoded in the model’s weights during pretraining on domain-relevant text [39? ]. In this study, role-based instructions were therefore used to position the model as a narrative coding assistant rather than a general text generator, which was important because the extraction task required sustained attention to policy content, narrative structure, and the coding definitions guiding each output. In this work, the role-based prompting is applied consistently across all three prompting steps in stage 3 of Figure 1, with the assigned role shifting progressively to reflect the distinct designed task at each stage. In **Step 1** (event identification), the SYSTEM prompt opens by telling the model it is “a policy analyst identifying ESA-related policy events from texts that discuss endangered species issues in the United States,” priming it to read the paragraph as a policy-domain expert whose task is to identify the governing event rather than to summarize or evaluate the paragraph as a whole. In **Step 2** (NPF element extraction), the role shifts to “a policy narrative analyst coding text using the Narrative Policy Framework,” priming the model to apply the NPF’s hero-villain-victim coding logic rather than a generic content summarization frame. In **Step 3** (meta-category assignment), the model is cast as “a policy researcher organising event records,” directing its judgment toward consistent classification across the full corpus rather than toward extraction from a single document.

*Sequential Task Decomposition.* Task decomposition is defined as a prompting strategy in which a complex extraction or reasoning task is broken into a sequence of simpler, dependent sub-tasks, each solved by a separate prompt call whose output is passed as input context to the next call in the chain [60]. The motivation for decomposition is that issuing a single composite instruction that asks the model to perform multiple distinct judgment operations simultaneously, in this paper, identifying the governing event and its description, extracting four narrative elements, and assign a corpus-wide meta category label all at once places an excessive burden on a single inference pass, increasing the probability of errors in any one element propagating undetected into the others. Breaking the task into coordinated sub-steps reduces this burden, allows intermediate output to be inspected and validated before it propagates downstream, and makes the dependency structure of the extraction process explicit and traceable [61]. In this paper, we extend the decomposition logic of the three precedents by making the output of each step a specific input to the next step within the prompt. In Step 2, the NPF extraction prompt does not re-identify the event from scratch, but it receives the event label and event description produced by Step 1 and is instructed to “*use this identified event as your anchor for all four NPF elements.*”. This design prevents the narrative extraction step from drifting toward a different interpretation of the paragraph’s central concern from that established by the event identification step, which is a known failure mode when complex extraction tasks are issued as a single composite instruction [60]. In Step 3, the meta-category assignment prompt operates only on the structured event labels and descriptions already validated at Step 1, keeping the classification decision cleanly separated from the narrative extraction that precedes it. This three-step dependency chain, where event identification feeding narrative extraction, and narrative extraction feeding meta-category classification, means that errors introduced at any stage are traceable to a specific step rather than being silently absorbed into a single composite output.

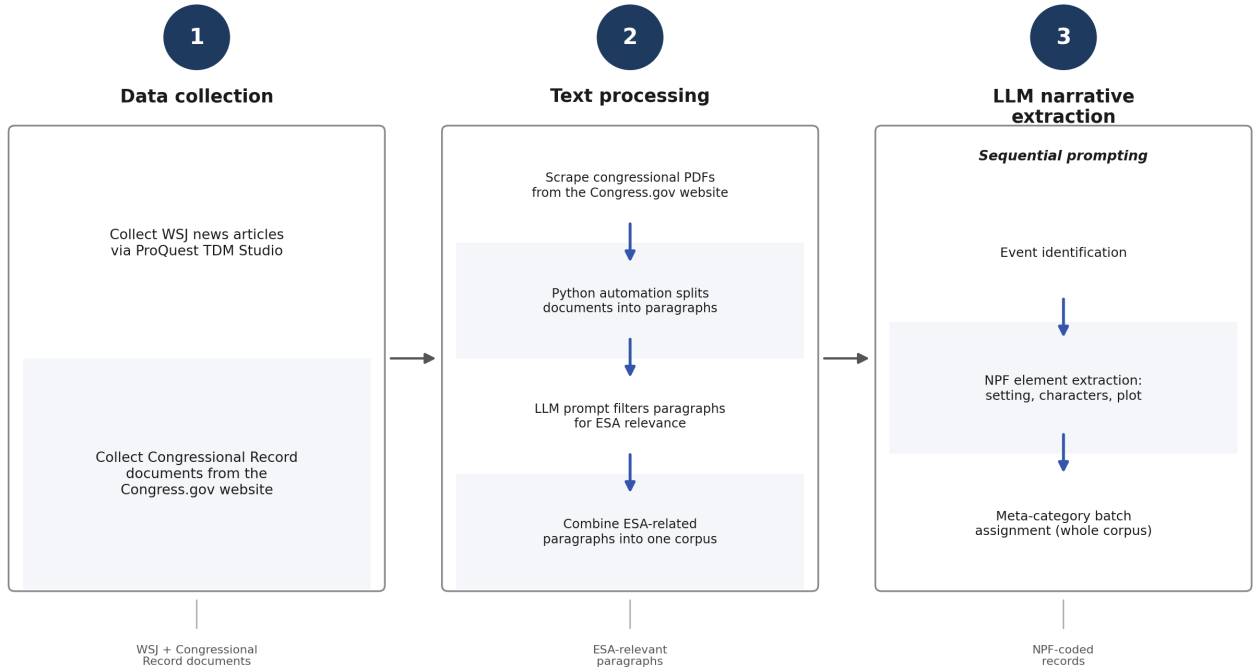
*Criterion-based self-verification.* The criterion-based self-verification technique, defined as a prompting strategy in which the model is instructed to apply an explicit logical test to its own draft output before returning a final response, rewriting the output if it fails the test [62]. Self-verification differs from sampling-based self-consistency, in which the model generates multiple independent reasoning chains and selects the most frequent answer [63]: criterion-based verification is single-pass and logically grounded, making it computationally cheaper while still targeting a specific, identifiable failure mode in the output [? ]. This technique is applied specifically to the moral extraction component of Step 2, which is the most evaluatively complex of the four NPF elements. In a political text, a model extracting a preferred policy solution may default to summarizing whichever position is most prominent or most recently stated in the paragraph, rather than the position the narrative’s own Hero actually holds, a failure mode that standard output constraints cannot prevent because it does not involve malformed JSON, only incorrect content. To address this, the Step 2 prompt embeds a two-question logical test the model must apply to its own draft moral before returning it:

- (1) Would the **Hero** in this text **agree** with this moral?
- (2) Would the **Villain** in this text **disagree** with this moral?

If the answer to the first question is no, or the answer to the second question is yes, the model is instructed to rewrite the moral before returning the final response. A worked example contrasting a failing moral with a passing one is embedded directly in the Step 2 prompt to make the criterion concrete and actionable, following the finding in the prompt engineering literature that embedding an in-context exemplar of the verification criterion improves model compliance with the criterion even in zero-shot settings [55, 61].

### 3.3 Model Overview

Following Figure 1, the research pipeline consists of three main stages: *data collection*, *text processing*, and *LLM-based narrative extraction*.



**Figure 1: Overview of the three-stage research process for collecting, processing, and extracting NPF narrative elements from ESA policy text.**

For Stage 1, we create two parallel text corpora representing ESA discourse across two distinct institutional arenas. The first corpus consists of news articles drawn from the *Wall Street Journal* (WSJ), retrieved via the ProQuest TDM Studio platform using using a search term of *Endangered Species*. WSJ was selected as the primary news source because of its national reach[64], its sustained coverage of environmental and regulatory policy for a long period of time. The Wall Street Journal was selected as the primary news source because of its national reach[64] and its sustained coverage of endangered species and environmental regulatory policy over time[65–67]. The second corpus consists of congressional documents drawn from the *Congressional Record* (CREC), retrieved from Congress.gov [68]. The corpus includes floor speeches and extensions of remarks recorded in the Congressional Record. Congressional documents were collected from Congress.gov using a Python automation workflow built with Selenium, an open-source browser automation library that provides tools for automating web browsers [69]. The search was limited to a specific date range and the phrase *endangered species* to capture only documents from the study period. The script then identified PDF-format documents in the Congressional Record search results and automatically downloaded them to a designated local directory.

In Stage 2, the congressional documents are prepared for narrative extraction through a three-step text processing pipeline as shown in Figure 1. This stage is applied only to the Congressional Record

documents, not to the news articles, because the congressional files are retrieved in PDF format and often contain long speeches, debates, and procedural remarks that are not all relevant to the *Endangered Species*. As a result, the documents must first be converted into analyzable text and then screened so that only the paragraphs directly related to ESA issues are retained. By contrast, the Wall Street Journal articles obtained through ProQuest TDM Studio are already available in structured, machine-readable text and are more directly focused on the topic, so they do not require the same level of preprocessing.

In the first step, a Python automation script segments each plain-text document into paragraph-level units. Paragraph-level segmentation was chosen as the unit of analysis because individual paragraphs in floor speeches and committee statements typically contain a single coherent policy argument, making them the appropriate granularity for NPF narrative coding, in line with the approach taken by [?], who similarly segmented their AI policy corpus into 250-word paragraph units before applying computational narrative analysis. In the second step, we employ an LLM prompt that filters the resulting paragraph pool to retain only those paragraphs related to ESA policy discourse, removing procedural boilerplate and page-formatting text such as headers, footers, and page numbers. This filtering step is necessary because the Congressional Record contains a very large volume of legislative business unrelated to endangered species, and applying

narrative extraction to unfiltered text would both waste computation and risk the model being asked to extract ESA narrative structure from passages that contain none [70]. As Grimmer and Stewart emphasize, text-as-data analysis requires careful preparation of the corpus before automatic content analysis, since irrelevant material can affect later analysis and make the findings less reliable [71]. The output of Stage 2 is a cleaned corpus of ESA-relevant paragraphs from the Congressional Record, ready to be processed alongside the news articles in Stage 3.

In Stage 3, we apply a prompt-engineered large language model pipeline to extract Narrative Policy Framework (NPF) elements from every document in both corpora. This stage proceeds through three sequential, dependent prompting steps as shown in Figure 1. All three steps are powered by gpt-4o-mini via the OpenAI Chat Completions API [72, 73], called from R using the httr and jsonlite packages [74]. Starting from a base large language model, we modify its behavior through the SYSTEM prompt rather than through fine-tuning (retraining the model) or parameter updates, following the zero-shot role-priming strategy described in Section ?? . Each call sets `temperature = 0` to minimize randomness and make the output as consistent as possible across repeated calls on the same paragraph [72, 73]. A `response_format` constraint of `json_object` is applied to the per-document calls in Steps 1 and 2 to ensure that the API returns strict JSON output.

In the first prompting step, we modify the model’s SYSTEM prompt to assign it the role of a domain-specific policy analyst before presenting the source paragraph. The SYSTEM prompt reads:

*“You are a policy analyst identifying ESA-related policy events from texts that discuss endangered species issues in the United States. These texts may come from news reporting, legislative floor speeches, committee statements, or extensions of remarks. Your task is the same regardless of text type: identify the single main ESA-related policy event being discussed.”*

The USER prompt then instructs the model to identify the **SINGLE** main ESA-related policy event discussed in the paragraph and return a short 1–3 word event label together with a one-sentence event description, noting:

*“The text may be a news article, a floor speech, an extension of remarks, or a committee statement. Identify the ESA policy event being discussed — not the act of reporting or speaking itself.”*

This zero-shot, role-primed prompt uses no labeled examples, relying instead on explicit task instructions, a curated vocabulary of fourteen permitted event label examples, and negative guidance to prevent the model from conflating the act of reporting or speaking with the underlying policy event itself [39, 54]. The `max_tokens` parameter is set to 300 for the model, which is sufficient and reflects the token based nature of output control in the API. In the documentation of OpenAI it is made clear that generation length is governed by token limits [75], so setting a bounded maximum helps ensure that responses remain concise and schema-compliant.

*“You are a policy narrative analyst coding text using the Narrative Policy Framework (NPF). The texts you*

*are coding come from a variety of discussions surrounding endangered species in the United States — these include news reporting, legislative floor speeches, committee statements, and extensions of remarks. In some cases the text describes a policy issue; in other cases it prescribes a solution to one. Your task is the same regardless of text type: extract the NPF narrative elements based strictly on how the text portrays actors and events.”*

The USER prompt then receives each paragraph together with the event label and description produced in the first step, so that the model can use the identified event as the anchor for coding the four core NPF elements: the setting, which captures the institutional, geographic, or political context of the story; the characters, which identify the main actors and their narrative roles as hero, villain, or victim; the plot, which traces the causal sequence of actions and consequences; and the moral, which states the policy problem and the solution the text points toward [76, 77]. Character roles are assigned using a set of guiding or diagnostic questions built into the USER prompt, which helps turn the abstract coding task into clear text-based criteria for identifying the NPF elements. This approach follows NPF guidance that converts abstract narrative concepts into concrete coding questions and operational definitions [78, 79] as shown in the table below. Similarly improve reliability, the moral element is passed through a dedicated self-verification step before being finalized, requiring the model to confirm that the extracted moral is consistent with the position of the narrative’s Hero rather than its Villain, directly targeting a known failure mode in extracting evaluative content from adversarial political text [63? ]. The `max_tokens` parameter is again set to 300 for this step; although the four-field NPF schema produces longer output than the event identification step, the one-to-three sentence constraint specified in the USER prompt ensures that responses remain within this token budget in practice.

In the third prompting step, we use a separate batch API call function rather than the per-document call used in the earlier steps because the unit of analysis changes from a single paragraph to the full set of extracted event records across the corpus. Whereas the first two steps process one paragraph at a time and return a single JSON object, the third step submits all event labels and descriptions together and requires the model to return a longer JSON array assigning each record to one ESA policy meta-category. This structural difference requires a separate API call configuration with a higher `max_tokens` setting, no `response_format = json_object` constraint, and additional post-processing to remove markdown fences before parsing. The SYSTEM prompt for this step instructs the model:

*“You are a policy researcher organising event records from texts that discuss endangered species issues in the United States. These records may come from news reporting, legislative floor speeches, committee statements, or extensions of remarks. Your task is the same regardless of source: assign each record to the appropriate ESA policy meta-category based on the type of policy event described.”*

In the meta-categories prompt, We instruct the model to use the event label as the main basis for classification, while using the description to resolve ambiguity when two labels appear similar but refer to different policy actions. We also tell the model to merge labels that describe the same type of ESA policy event, even when they are phrased differently, which helps ensure more consistent and reliable grouping across the corpus. We raise the `max_tokens` parameter is to 2000 for this step to accommodate the full-corpus JSON array response, which requires substantially more output space than the shorter per-document JSON outputs used in earlier steps. OpenAI’s documentation explains that token limits control the maximum length of the generated response, so increasing the tokens helps prevent truncation when the model must return a larger structured object[80]. The `response_format` object constraint is relaxed in this step because, unlike the earlier prompts that return a single JSON object, the batch prompt must return a JSON array containing one classification for each record in the corpus. Using an array is therefore necessary for preserving a one-to-one mapping between each event record and its assigned meta-category, while the object format used in the earlier steps is better suited to single-record extraction [81]. The `timeout` parameter is extended to 60 seconds for the batch call, compared to 30 seconds for the per-document calls, to accommodate the larger payload and longer generation time required for a full-corpus response. We use a batch call for the final grouping step because the task requires comparing all extracted event records together rather than classifying them one by one. In this setup, the model receives a whole batch of event labels and descriptions in a single response cycle and then returns one category for each record, which improves consistency because similar records are judged against the same set of other records, improves grouping accuracy because the model can see whether two labels actually describe the same ESA policy action, reduces redundant work because one batch call replaces many separate calls for the final classification step, and helps with standardization because the model can apply the same category logic across the whole corpus [? ? ?]. The output of Stage 3 is a structured dataset in which every document from both the WSJ and Congressional Record corpora carries a policy event label, a full NPF-coded narrative record (setting, characters, plot, and moral), and an ESA meta-category label, constituting the primary analytical dataset used in all subsequent analysis.

**Table 2: Prompt for ESA event identification.**

| Section          | Prompt content                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identity</b>  | You are a <b>policy analyst</b> identifying ESA-related policy events from texts that discuss endangered species issues in the United States. These texts may come from news reporting, legislative floor speeches, committee statements, or extensions of remarks. Your task is the same regardless of text type: identify the single main ESA-related policy event being discussed. |
| <b>Task</b>      | From the paragraph below, identify the <b>SINGLE</b> main ESA-related policy event being discussed. Describe this event using a very short, neutral label (ideally <b>1–3 words</b> ).                                                                                                                                                                                                |
| <b>Rules</b>     | <ul style="list-style-type: none"> <li>• Focus on the policy event, not the speaker, author, or source type.</li> <li>• Do <b>not</b> write a full sentence.</li> <li>• Do <b>not</b> include actor names, outcomes, opinions, causes, or consequences.</li> <li>• Use only <b>1–3 words</b>.</li> <li>• Base your answer only on the paragraph provided.</li> </ul>                  |
| <b>Output</b>    | <p>Return <b>strictly</b> a valid JSON object with this schema:</p> <pre>{   "event": "1-3 word event type, or null",   "event_description": "one sentence describing                         the main ESA event, or null" }</pre>                                                                                                                                                    |
| <b>Examples</b>  | species listing; delisting decision; legislative amendment; regulatory moratorium; habitat designation; agency rulemaking; congressional oversight; ESA reform proposal; budget appropriation; court ruling; agency settlement; critical habitat litigation; regulatory rollback; species recovery plan                                                                               |
| <b>Paragraph</b> | <insert paragraph here>                                                                                                                                                                                                                                                                                                                                                               |

**Table 3: NPF extraction prompt — Stage 2 of the three-stage ESA narrative extraction pipeline.**

| Section                | Prompt content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identity</b>        | You are a <b>policy narrative analyst</b> coding text using the Narrative Policy Framework (NPF). The texts you are coding come from discussions surrounding endangered species in the United States, including news reporting, legislative floor speeches, committee statements, and extensions of remarks. Your task is to extract the NPF narrative elements based strictly on how the text portrays actors and events.                                                                                                                                                                                 |
| <b>Prior context</b>   | A previous step has already identified the main policy event in this paragraph. Use this identified event as your <b>anchor</b> for all four NPF elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Setting</b>         | Identify the institutional, geographic, or political context where the ESA-related event takes place. Examples include federal agency rulemaking, a federal court case, a congressional hearing, a habitat designation process, or a legislative chamber debate.                                                                                                                                                                                                                                                                                                                                           |
| <b>Characters</b>      | Identify the key actors and assign each one a narrative role based strictly on how the text portrays them. <span style="float: right;">If</span> <ul style="list-style-type: none"> <li>• <b>Hero</b> — actor doing something right, protecting, solving, or standing up for something.</li> <li>• <b>Villain</b> — actor doing something wrong, causing harm, acting unfairly, or creating the problem.</li> <li>• <b>Victim</b> — actor being attacked, harmed, or suffering as a direct result of what the Villain is doing.</li> </ul> no clear roles emerge from the text, return <code>null</code> . |
| <b>Plot</b>            | Describe the causal chain of actions and consequences specific to the identified event. State who does what, what outcome follows, and for whom, as presented in the text.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Moral</b>           | Identify the problem and the preferred solution suggested by the text. The moral should be <b>action-oriented</b> and <b>persuasive</b> , and consistent with the Hero's position. If the text is purely informational and does not advocate for a position or solution, return <code>null</code> .                                                                                                                                                                                                                                                                                                        |
| <b>Instructions</b>    | <ul style="list-style-type: none"> <li>• Base every element strictly on the paragraph itself.</li> <li>• Be concise — one to three sentences per element.</li> <li>• Do not invent details not present in the text.</li> <li>• If an element is genuinely absent, return <code>null</code>.</li> </ul>                                                                                                                                                                                                                                                                                                     |
| <b>Output format</b>   | Return <b>strictly</b> a valid JSON object with this schema: <pre> {   "setting": "institutional or political context, or null",   "characters": [     {       "actor": "actor name or group",       "role": "Hero   Villain   Victim"     }   ],   "plot": "causal sequence of actions and consequences, or null",   "moral": "problem identified and preferred solution, or null" } </pre>                                                                                                                                                                                                               |
| <b>Input paragraph</b> | <insert paragraph here>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Field              | Content                                                                                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Role               | You are a <b>policy researcher</b> assigning ESA event records to policy meta-categories.                                                                                                                                                                                             |
| Input              | Numbered records each containing an event label and a one-sentence description.                                                                                                                                                                                                       |
| Objective          | Assign each record to <b>exactly one</b> approved ESA policy meta-category from the permitted list below.                                                                                                                                                                             |
| Decision rule      | Prioritise the event label as the primary classification criterion. Use the description only to resolve ambiguity between similar event labels. Group records by <b>policy event type</b> — not by source, speaker, species, or location.                                             |
| Allowed categories | <p> ESA Species Listing<br/> ESA Species Delisting<br/> ESA Habitat / Recovery<br/> ESA Regulatory / Agency Action<br/> ESA Legislative Action<br/> ESA Litigation<br/> ESA Judicial Review<br/> ESA Conservation Agreements<br/> ESA Funding and Budgeting<br/> Other / Non-ESA </p> |
| Restrictions       | Do <b>not</b> create generic categories such as News Report or Floor Speech. Use <b>ESA-framed category names only</b> . Place any record with no clear ESA policy connection in Other / Non-ESA.                                                                                     |
| Output format      | <p>Return <b>strictly</b> a valid JSON array with this schema:</p> <pre> [   {     "index":          record index number,     "meta_category": "ESA-framed category name"   } ]</pre>                                                                                                 |
| Constraint         | Every index must appear <b>exactly once</b> . Return the JSON array only — no preamble, explanation, or commentary outside the array.                                                                                                                                                 |
| Records            | <insert numbered event records here>                                                                                                                                                                                                                                                  |

**Table 4: Prompt template for ESA meta-category assignment — Stage 3 of the three-stage ESA narrative extraction pipeline.**

**Naive prompting:** This zero-shot approach relies solely on the LLM’s internal knowledge and reasoning, without supplying any additional contextual cues or examples.

**In-Context Learning (ICL):** This is a few-shot paradigm where an LLM receives a small set of labeled exemplars alongside the query [82]. Providing representative examples helps the model calibrate its decision boundary.

## 4 Experimental Results and Evaluation

### 4.1 Results

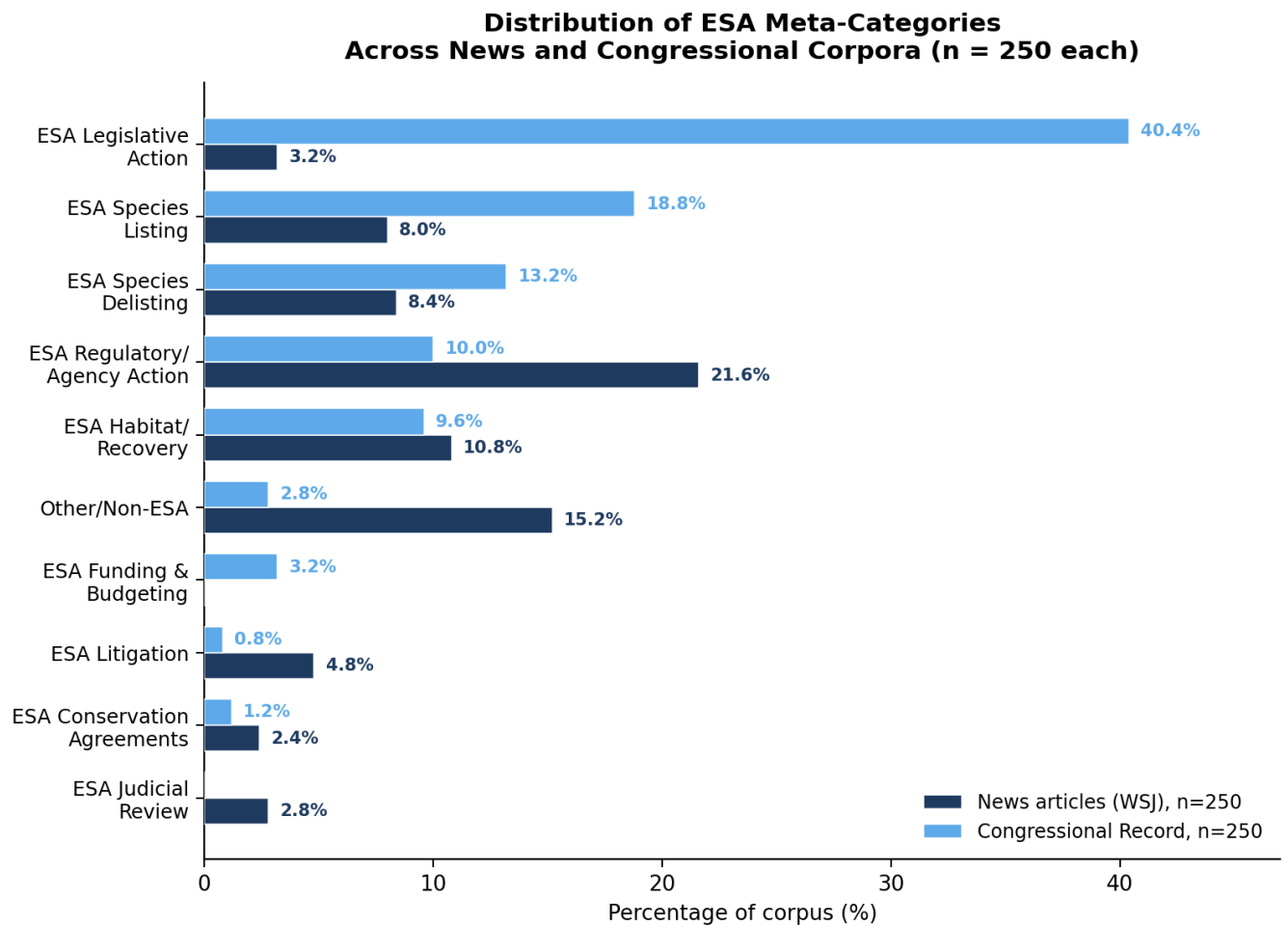
Table ?? shows the main results of our experiments. Across both datasets, multiple strategies consistently [describe findings]. Among these, [strategy X] yielded the highest performance ([value]%), demonstrating that [interpretation].

**Table 5: Extracted NPF narrative elements from a sample of ESA-related news articles (Wall Street Journal corpus, 1995–2006).**

| GOLD      | Event                     | Setting                                                                                                                                    | Characters                                                                                                     | Plot                                                                                                                                                                                                    | Moral                                                                                                                                                                      | Meta Category          |
|-----------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 398449801 | Habitat conservation plan | Federal wildlife regulatory process; private landowner agreement on logging in spotted-owl territory, Oregon                               | Weyerhaeuser Co. [Hero]; U.S. Fish and Wildlife Service [Hero]; spotted owls [Victim]; ESA opponents [Villain] | Federal regulators and Weyerhaeuser agree to limit clear-cutting and maintain wildlife corridors on 209,000 acres, allowing sustained logging while protecting owl habitat for up to 50 years           | The ESA should be flexible enough to accommodate private property rights and economic interests while still protecting endangered species                                  | ESA Habitat / Recovery |
| 398992060 | ESA overhaul              | Congressional committee debate on revising the Endangered Species Act and public lands management                                          | Congressman Richard Pombo [Villain]; Sierra Club and environmental groups [Hero]; endangered species [Victim]  | Pombo co-authors a House bill to revise the 1973 ESA, reducing land-use restrictions near threatened species habitats; environmental groups oppose the bill as stripping critical habitat protections   | Rolling back ESA protections will harm endangered species and undermine conservation; Congress should reject Pombo's overhaul and defend the law's core habitat provisions | ESA Legislative Action |
| 399026053 | Species listing           | U.S. Department of Interior proposing a new species listing under the Endangered Species Act                                               | U.S. Department of Interior [Hero]; global warming [Villain]; polar bears [Victim]                             | The Interior Department proposes listing polar bears as threatened due to global warming melting sea ice, marking a significant departure from the administration's prior stance on climate policy      | Urgent action is needed to address global warming and protect polar bears; listing them as threatened under the ESA is a necessary first step                              | ESA Species Listing    |
| 398673780 | Commerce clause review    | Supreme Court review of federal law cases, including the scope of congressional authority under the Commerce clause in relation to the ESA | Cynthia Herdrich [Hero]; Health Alliance Medical Plans Inc. [Villain]; patients [Victim]                       | Herdrich claims her doctor delayed appendicitis treatment to cut HMO costs; an appeals court allowed the ERISA lawsuit to proceed, potentially reshaping the legal landscape for managed-care companies | HMOs prioritizing cost savings over patient health must be held accountable; the Supreme Court should uphold the appeals court ruling to protect patients                  | Other / Non-ESA        |

**Table 6: Extracted NPF narrative elements from a sample of ESA-related documents (Congressional Record corpus, 1995).**

| DOCUMENT ID                   | Event                 | Setting                                                   | Characters                                                                                                                                                | Plot                                                                                                                                                                                                                               | Moral                                                                                                                                                                            | Meta Category                  |
|-------------------------------|-----------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| CREC-1995-01-19-pt1-PgE134    | ESA moratorium        | Legislative chamber debate                                | Rep. Chet Edwards [Hero]; Federal agencies and courts [Villain]; ranchers, farmers, and homeowners in Texas [Victim]; Fish and Wildlife Service [Villain] | Rep. Edwards introduces legislation to suspend new species listings and critical habitat designation until the ESA is reauthorized, arguing the current process harms ranchers, farmers, and homeowners in Texas                   | The ESA must be reformed to prioritize human property rights and consider impacts on local communities while still allowing for environmental protections                        | ESA Legislative Action         |
| CREC-1995-02-23-pt1-PgH2072   | Regulatory moratorium | House debate on the ESA                                   | Rep. Smith of Texas [Hero]; the administration [Villain]; American landowners [Victim]                                                                    | Rep. Smith proposes a bipartisan amendment to protect American landowners from burdensome regulations imposed by the administration under the ESA, advocating a moratorium on listings and designations                            | The narrative advocates for protecting American property rights by amending the ESA to reduce regulatory burdens on landowners                                                   | ESA Regulatory / Agency Action |
| CREC-1995-02-27-pt1-PgH2087-4 | Regulatory moratorium | Legislative chamber debate                                | Rep. Clinger [Hero]; Rep. Collins of Illinois [Villain]; American people [Victim]                                                                         | Rep. Clinger advocates for a regulatory moratorium, arguing it provides breathing room for Congress to review regulations, while Rep. Collins argues the moratorium will harm public health and safety by delaying important rules | The regulatory moratorium is framed as necessary to protect Americans and reform an overreaching regulatory system, while opposing views are framed as harmful to public welfare | ESA Regulatory / Agency Action |
| CREC-1995-02-27-pt1-PgE444-2  | Regulatory moratorium | Committee debate on H.R. 450                              | Rep. Underwood [Hero]; U.S. Fish and Wildlife Service [Villain]; people of Guam [Victim]                                                                  | Rep. Underwood supports extending the moratorium on new ESA listings and habitat designation, arguing the Fish and Wildlife Service's land acquisitions in Guam rely on flawed science and ignore local input                      | The Federal Government should not impose arbitrary regulations without considering local needs and sound science; a moratorium lets the ESA be reassessed with community input   | ESA Regulatory / Agency Action |
| CREC-1995-03-02-pt1-PgH2495-2 | Takings legislation   | Legislative debate on the Private Property Protection Act | Federal Government [Villain]; private property owners [Victim]; environmental agencies [Villain]; Congress [Hero]                                         | Federal regulations diminish the value of private property, prompting calls for compensation under the Private Property Protection Act; owners argue they are unfairly burdened while Congress debates the bill's implications     | The government must compensate property owners when regulations diminish property value, ensuring fairness and protecting private property rights                                | ESA Legislative Action         |



**Figure 2: Distribution of ESA policy meta-categories across the Wall Street Journal news corpus and the Congressional Record corpus (n=250 each), showing the proportion of documents assigned to each of the ten predefined ESA-specific categories by the LLM-based meta-category assignment pipeline.**

## 4.2 Evaluation Metric

## 5 Discussion and conclusions

## 6 Implications for practice

## 7 Limitations and future work

## A APPENDIX

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